

Surname / Van:

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FSK 176**Semester Test 2 - Semestertoets 2**

Department of Physics, University of Pretoria
Departement Fisika, Universiteit van Pretoria

Date / Datum: 12/10/2017

Time / Tyd: 2.5 hours / ure

Total / Totaal : 70 max / maks

Examiner / Eksaminator: Dr JM Nel

Moderator / Moderator: RQ Odendaal

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Test register: Tear carefully on the line above and hand to the invigilator.

Toets register: Skeur versigtig op die lyn hierbo en handig dit in by die toesighouer.

FSK 176: Semester Test 2 / Semestertoets 2**12/10/2017**

Surname Van					Initials Voorletters		
Student number Studentenommer							
Venue Lokaal							
Signature Handtekening							

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Please read the following carefully. *Lees die volgende noukeurig*

1. Answer all the questions in the space provided on the question paper. *Antwoord alle vrae in die spasie gelaat op die vraestel.*
2. No pencil work will be marked. Do not use "tippex". *Geen potlood-werk sal gemerk word nie. Moet nie uitwis-vloeistof ("tippex") gebruik nie.*
3. If you need more space, please use the back of the pages, and clearly indicate where the question continues. *As u meer spasie benodig gebruik die agterkant van die bladsye. Dui duidelik aan waar die vraag aangaan.*
4. Where applicable you need to draw the necessary diagrams (e.g. free-body diagrams). *Teken diagramme waar nodig (bv. vryliggaams-diagramme).*
5. Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant physics. **Only substitute numerical values in the final step of calculations! A correct answer does not imply that you will receive full marks.** *U antwoorde moet sistematies en logies neergeskryf word. U moet bewys dat u die nodige fisika verstaan en kan toepas. Numeriese waardes mag slegs in die laaste stap van 'n berekening vervang word! 'n Korrekte antwoord beteken nie dat u noodwendig vol punte sal kry nie.*
6. Units must be used when numbers are substituted else marks will be subtracted. *Eenhede moet gebruik word wanneer numeriese waardes vervang word, andersins sal punte afgetrek word.*
7. The use of the cheat sheet is only allowed subject to the rules stated on it. *Die formuleblad mag slegs gebruik word onderhewig aan die reëls daarop gedruk.*
8. All exam regulations of the University of Pretoria apply, and any contravention of it will be treated in a serious manner. *Alle eksamen regulasies van die Universiteit van Pretoria geld en enige oortreding daarvan sal in 'n ernstige lig beskou word.*

Constants and formulas / Konstantes en formules

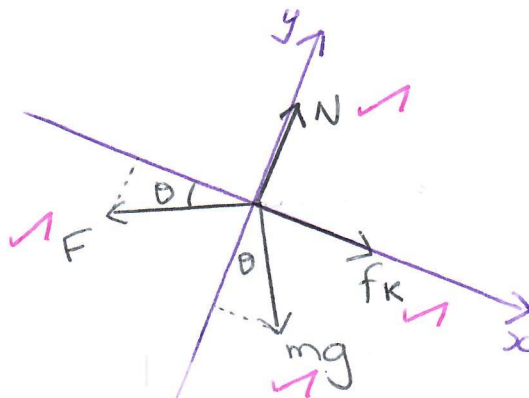
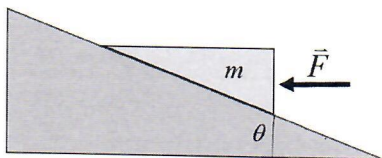
$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$; $G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$; $e = 1.602 \times 10^{-19} \text{ C}$; $k = 1.381 \times 10^{-23} \text{ J/K}$; $m_e = 9.109 \times 10^{-31} \text{ kg}$; $m_p = 1.673 \times 10^{-27} \text{ kg}$; $R = 8.314 \text{ J}/(\text{mol}\cdot\text{K})$; $N_A = 6.02 \times 10^{23} / \text{mol}$; $1 \text{ atm} = 101.3 \text{ kPa}$; $1 \text{ u} = 1.6605677 \times 10^{-27} \text{ kg}$; $c = 3 \times 10^8 \text{ m/s}$; $V_{\text{sphere/sfeer}} = \frac{4}{3}\pi r^3$; $F_{\text{gravity}} = \frac{Gm_1m_2}{r^2}$; $1 \text{ litre} = 1000 \text{ cm}^3$; $M_{\text{Earth}} = M_{\text{Aarde}} = 5.97 \times 10^{24} \text{ kg}$; $\rho(\text{water}) = 1 \text{ g/cm}^3$; $r_{\text{Earth}} = r_{\text{Aarde}} = 6.37 \times 10^6 \text{ m}$

1 Question 1 / Vraag 1

[6]

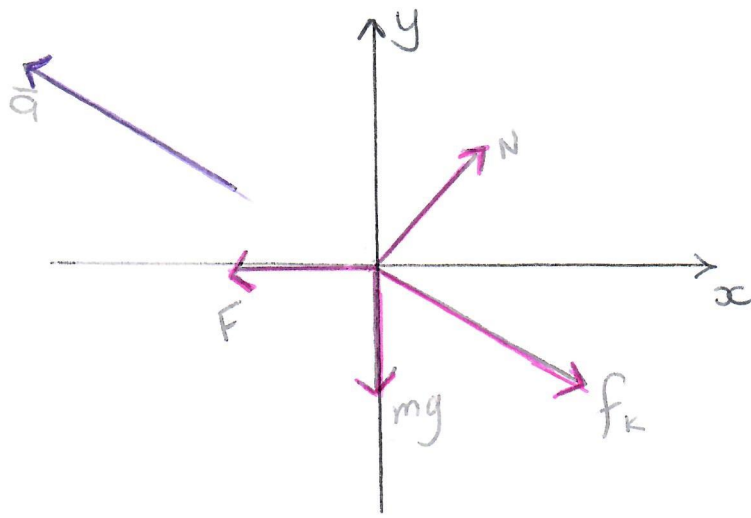
A wedge of mass $m = 36.1 \text{ kg}$ lies on a plane that is inclined by an angle $\theta = 21.3^\circ$ with respect to the horizontal. A force $F = 302.3 \text{ N}$ in the horizontal direction pushes on the wedge, as shown in the figure. The coefficient of kinetic friction between the wedge and the plane is 0.159. Determine the acceleration of the wedge along the plane.

'n Wig met massa $m = 36.1 \text{ kg}$ lê op 'n skuinsvlak wat teen 'n hoek $\theta = 21.3^\circ$ gekantel is ten opsigte van die horisontaal. 'n Horisontale krag $F = 302.3 \text{ N}$ stoot die wig soos aangedui in die figuur. Die kinetiese wrywingskoëffisiënt tussen die wig en die vlak is 0.159. Bepaal die versnelling van die wig langs die skuinsvlak.



No axes

(-1)



Question 1

Alternate set of axes.

Using this reference frame, you will, however, need to determine a_x and a_y first and then determine $\vec{a}$

$$\Sigma F_y = ma_y = 0 = N - mg \cos \theta - F \sin \theta$$

$$N = mg \cos \theta + F \sin \theta \quad \checkmark$$

$$\Sigma F_x = ma_x = f_k - F \cos \theta + mg \sin \theta$$

$$= \mu N - F \cos \theta + mg \sin \theta \quad \checkmark$$

$$a_x = \frac{\mu (mg \cos \theta + F \sin \theta) - F \cos \theta + mg \sin \theta}{m} \quad \checkmark$$

$$= 0.159 (36.1 \text{ kg} (9.81 \text{ m/s}^2 \cos 21.3^\circ + 302.3 \text{ N} \sin 21.3^\circ) -$$

$$\frac{302.3 \text{ N} \cos 21.3 + 36.1 \text{ kg} (9.81 \text{ m/s}^2 \sin 21.3^\circ)}{36.1 \text{ kg}}$$

$$a = -2.305 \text{ m/s}^2 \quad \checkmark$$

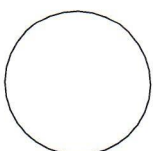
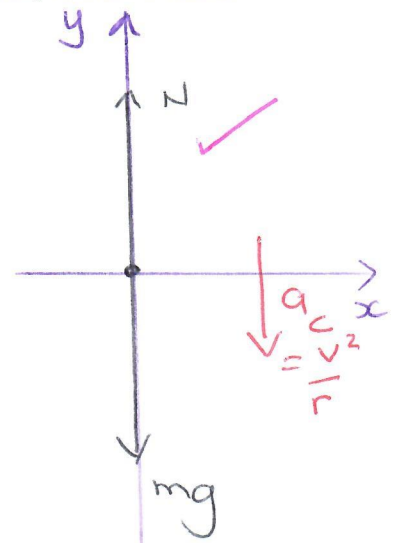
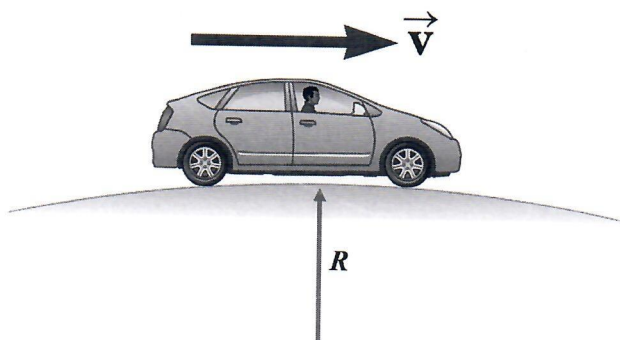
$$a = 2.305 \text{ m/s}^2 \text{ up the slope} \quad \checkmark$$

2 Question 2 / Vraag 2

[10]

Disturbed by speeding cars outside his house, a FSK176 student designed a speed bump and had it installed. Suppose a 1600 kg car passes over the bump in a road that follows the arc of a circle with a radius of $R = 20.5 \text{ m}$ as shown in the figure.

Gepla deur karre wat by sy huis verby jaag, het 'n FSK176 student 'n spoedwal ontwerp en laat installeer. Veronderstel dat 'n 1600 kg kar oor die wal in die pad ry wat 'n sirkelboog is met 'n radius van $R = 20.5 \text{ m}$, soos aangedui in die figuur.



- (a) When the car travels at 30.0 km/h, what force does the road exert on the car as it passes the highest point of the bump?

Wanneer die kar teen 30.0 km/h oor die spoedwal ry, wat is die krag wat die pad op die kar uitoeft soos dit oor die hoogste punt van die wal beweeg? (5)

$$v = 30 \frac{\text{km}}{\text{h}} \times \frac{1000\text{m}}{1\text{km}} \times \frac{1\text{hr}}{60\text{min}} \times \frac{1\text{min}}{60\text{s}} \checkmark$$

$$= 8.33 \text{ m/s} \checkmark$$

$$\sum F_y = -ma_y = N - mg = -m \frac{v^2}{r} \checkmark$$

$$N = -m \left(-\frac{v^2}{r} + g \right) \checkmark$$

$$= -1600 \text{ kg} \left(-\frac{(8.33 \text{ m/s})^2}{20.5 \text{ m}} + 9.8 \text{ m/s}^2 \right)$$

$$= 10.3 \times 10^4 \text{ N upwards} \checkmark$$

- (b) What is the maximum speed (in km/h) the car can have without losing contact with the road as it passes over the highest point?

Wat is die maksimum spoed (in km/uur) wat die kar kan hê sonder dat dit kontak verloor met die pad soos dit oor die hoogste punt van die spoedwal beweeg? (5)

Maximum speed will be when $N = 0 \text{ N}$

$$\therefore \sum F_y = m \frac{v^2}{r} = N - mg \checkmark$$

$$v = \sqrt{\frac{(N - mg)r}{m}} \checkmark$$

$$v = \sqrt{gr} \checkmark \text{ if } N = 0 \text{ N} \checkmark$$

$$= \sqrt{9.8 \text{ m/s}^2 (20.5 \text{ m})}$$

$$= 14.2 \text{ m/s} \left[\times \frac{1\text{km}}{1000\text{m}} \times \frac{60\text{s}}{1\text{min}} \times \frac{60\text{min}}{1\text{hr}} \right] \checkmark$$

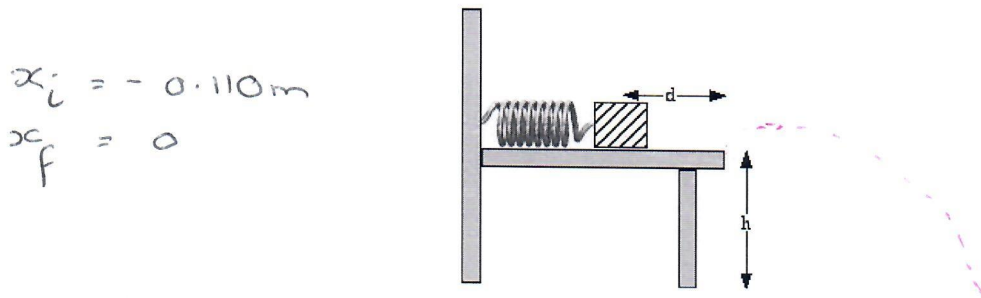
$$= 51.0 \text{ km/h} \checkmark$$

3 Question 3 / Vraag 3

[9]

Consider a block on a table. This block is pushed by a spring attached to the wall, and slides across the table. The block has a mass of 1.35 kg, the spring constant is $k = 560 \text{ N/m}$ and the spring is initially compressed by 0.110 m. The block lies a distance $d = 0.650 \text{ m}$ from the edge of the table when the spring is compressed. The table has a height of $h = 0.750 \text{ m}$. The coefficient of kinetic friction between the block and the table is $\mu_k = 0.160$.

Beskou 'n blok wat op 'n tafel lê. Die blok word deur 'n veer, wat vasgemaak is aan die muur, gedruk en dit gly oor die tafel. Die blok het 'n massa van 1.35 kg, die veerkonstante is $k = 560 \text{ N/m}$ en die veer is aanvanklik met 0.110 m saamgepers. Die blok lê 'n afstand $d = 0.650 \text{ m}$ van die rant van die tafel wanneer die veer saamgepers is. Die tafel het 'n hoogte van $h = 0.750 \text{ m}$. Die kinetiese wrywingskoëffisiënt tussen die blok en die tafel is $\mu_k = 0.160$.



- (a) Show clearly that the block will reach the end of the table with a speed of 1.73 m/s.

Wys duidelik dat die blok die rant van die tafel sal bereik teen 'n spoed van 1.73 m/s.

(4)

$$\Delta K + \Delta U_g + \Delta U_s + \Delta E_{\text{int}} = 0$$

$$\left(\frac{1}{2}mv_f^2 - 0\right) + (0 - 0) + \left(\frac{1}{2}kx_f^2 - \frac{1}{2}kx_i^2\right) + (\mu_k N d) = 0$$

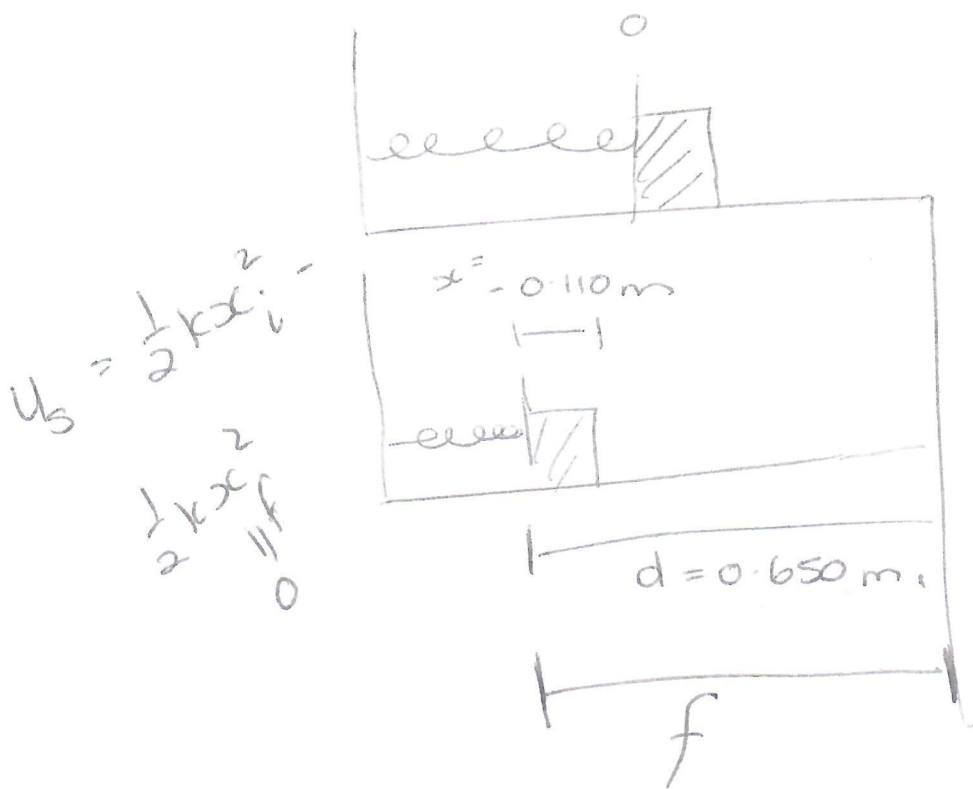
$$\frac{1}{2}mv_f^2 - \frac{1}{2}kx_i^2 + \mu_k mgd = 0$$

$$v_f^2 = \left(\frac{1}{2}kx_i^2 - \mu_k mgd\right) \cdot \frac{2}{m}$$

$$= \frac{(560 \text{ N/m})(0.110 \text{ m})^2 - (0.160)(1.35 \text{ kg})(9.8 \text{ m/s}^2)(0.650 \text{ m})}{1.35 \text{ kg}}$$

$$v_f = 1.73 \text{ m/s}$$

Question 3



Read the question carefully.

The spring is compressed by 0.110 m .

When compressed, the block is then 0.650 m from the edge of the table.

- (b) What speed will the block have when it lands on the floor?
 Wat sal die spoed van die blok wees wanneer dit op die vloer land?

(5)

$$\Delta E_{\text{mech}} = \Delta K + \Delta U_g + \Delta U_s = 0$$

$$\therefore \left(\frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 \right) + (mgh_f - mgh_i) + 0 = 0$$

Let Table height = 0 m and floor $h = -0.750 \text{ m}$

$v_i = 1.73 \text{ m/s}$ (from (a) at the edge of table)

$$\therefore v_f^2 = \left(\frac{1}{2} m v_i^2 - mgh_f + mgh_i \right) \frac{2}{m}$$

$$v_f = \left(v_i^2 - 2gh_f + 2gh_i \right)^{1/2}$$

$$= \left((1.73 \text{ m/s})^2 - 2(9.8 \text{ m/s}^2)(-0.750 \text{ m}) \right)^{1/2} = 4.21 \text{ m/s}$$

4 Question 4 / Vraag 4

[7]

A 7.80 g bullet moving at 575 m/s strikes the hand of a superhero, causing the hand to move 5.50 cm in the direction of the bullet's velocity before stopping.

'n 7.80 g koeël beweeg teen 575 m/s en tref dan die hand van 'n super-held. Die hand beweeg terug deur 'n afstand van 5.50 cm in die rigting van die koeël se snelheid voordat dit tot stilstand kom.

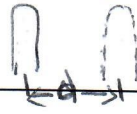
- (a) Use work and energy considerations to find the **average** force that stops the bullet.

Deur gebruik te maak van arbeid en energie konsepte vind die **gemiddelde** krag wat die koeël stop.

(4)

$$v_i = 575 \text{ m/s}$$

$$v_f = 0 \text{ m/s}$$



$$W = \Delta K$$

$$F_{\text{avg}} d \cos \theta = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$

$$F_{\text{avg}} = \frac{-\frac{1}{2} m v_i^2}{d \cos \theta}, \quad \theta = 180^\circ$$

$$= \frac{-\frac{1}{2} (7.8 \times 10^{-3} \text{ kg}) (575 \text{ m/s})^2}{(5.50 \times 10^{-2} \text{ m}) \cos 180^\circ}$$

$$= 23.4 \times 10^3 \text{ N}$$

in opposite direction to d

OR.

(b) Determine how long it takes for the hand to stop the bullet.

Bepaal hoe lank dit neem vir die hand om die koeël tot stilstand te bring.

(3)

$$\Delta p = I = F_{avg} \Delta t$$

$$\therefore m v_f - m v_i = F_{avg} \Delta t$$

$$\Delta t = \frac{m v_f - m v_i}{F_{avg}} = \frac{0 - 7.8 \times 10^{-3} \text{ kg} (575 \text{ m/s})}{23.4 \times 10^3 \text{ N}}$$

$$= 1.92 \times 10^{-4} \text{ s}$$

OR

$$\Delta x = \frac{1}{2} (v_{ix} + v_{fx}) t$$

$$t = \frac{2 \Delta x}{v_{ix} + v_{fx}} = 1.92 \times 10^{-4} \text{ s}$$

5 Question 5 / Vraag 5

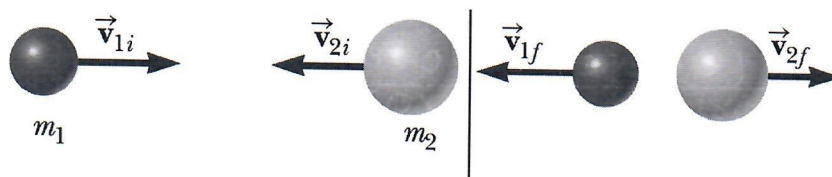
[10]

Consider two particles of masses m_1 and m_2 moving with initial velocities $\vec{v}_{1i}$ and $\vec{v}_{2i}$ along the straight line as shown in the figure (left). The two particles collide elastically head-on and then leave the collision site with different velocities, $\vec{v}_{1f}$ and $\vec{v}_{2f}$, as in the figure (right). Show clearly, with **logical explanations**, that the final velocity of m_1 is given by

$$v_{1f} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) v_{1i} + \left(\frac{2m_2}{m_1 + m_2} \right) v_{2i}.$$

Beskou twee deeltjies met massas m_1 en m_2 wat aanvanklik beweeg met snelhede $\vec{v}_{1i}$ en $\vec{v}_{2i}$ langs die reguitlyn soos aangedui in die figuur (links). Die twee deeltjies bots elasties kop-aan-kop en na die botsing beweeg hulle weg van die botsingspunt met verskillende snelhede, $\vec{v}_{1f}$ and $\vec{v}_{2f}$, soos aangedui in die figuur (regs). Wys duidelik, met **logiese verduidelikings**, dat die finale snelheid van m_1 gegee word deur

$$v_{1f} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) v_{1i} + \left(\frac{2m_2}{m_1 + m_2} \right) v_{2i}.$$



$$P_i = P_f$$

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f} \quad \text{--- (1)}$$

$$m_1 v_{1i} - m_1 v_{1f} = m_2 v_{2f} - m_2 v_{2i}$$

$$m_1 (v_{1i} - v_{1f}) = m_2 (v_{2f} - v_{2i}) \quad \text{--- (2)}$$

For an elastic collision:

$$K_i = K_f$$

$$\frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2 \quad \checkmark$$

$$m_1 v_{1i}^2 - m_1 v_{1f}^2 = m_2 v_{2f}^2 - m_2 v_{2i}^2$$

$$m_1 (v_{1i}^2 - v_{1f}^2) = m_2 (v_{2f}^2 - v_{2i}^2) \quad \checkmark$$

$$m_1 (v_{1i} - v_{1f})(v_{1i} + v_{1f}) = m_2 (v_{2f} - v_{2i})(v_{2f} + v_{2i}) \quad \checkmark \quad (3)$$

(3) $\div$ (2) gives $\checkmark$

$$v_{1i} + v_{1f} = v_{2f} + v_{2i}$$

$$v_{2f} = v_{1i} + v_{1f} - v_{2i} \quad \checkmark \quad (4)$$

(4) in (1) $\checkmark$

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 (v_{1i} + v_{1f} - v_{2i})$$

$$= m_1 v_{1f} + m_2 v_{1f} + m_2 v_{1i} - m_2 v_{2i} \quad \checkmark$$

Rearranging

$$(m_1 - m_2) v_{1i} + (m_2 + m_2) v_{2i} = (m_1 + m_2) v_{1f} \quad \checkmark$$

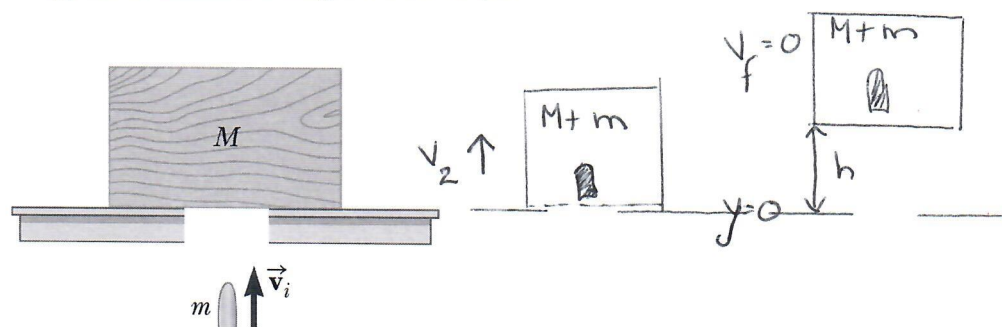
$$\therefore v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_{1i} + \frac{2m_2}{m_1 + m_2} v_{2i} \quad \checkmark$$

6 Question 6 / Vraag 6

[8]

A wooden block, of mass M , rests on a table over a large hole as shown in the figure. A bullet, with mass m and an initial velocity $\vec{v}_i$, is fired upward into the block and remains in the block after the collision. The block and bullet rise to a maximum height h above the table.

'n Hout blok, met massa M , lê op 'n tafel oor 'n groot opening, soos aangedui in die figuur. 'n Koeël, met massa m en 'n aanvanklike snelheid $\vec{v}_i$, word opwaarts in die blok gevuur en dit bly in die blok na die botsing. Die blok en die koeël styg tot 'n maksimum hoogte h bo die tafel.



- (a) Find an expression which gives the initial velocity of the bullet as a function of the maximum height, h , to which the block-bullet system rises.

Vind 'n vergelyking wat die aanvanklike snelheid van die koeël bepaal as 'n funksie van die maksimum hoogte, h , wat die blok-koeël stelsel bereik.

(6)

Initial Collision: $P_i = P_f$

$$mv_i + M(v_i) = (m + M)v_2$$

$$v_i = \frac{(m + M)v_2}{m}$$

Block-Bullet-Earth system: Conservation of E_{mech}

$$\Delta E_{\text{mech}} = 0 = \Delta K + \Delta U_g$$

Let table top be $y=0$, $U_g = 0$

$$\therefore \left(\frac{1}{2}(M+m)v_f^2 - \frac{1}{2}(M+m)v_2^2 \right) + ((M+m)gy_f - (M+m)gy_i) = 0$$

$$0 - \frac{1}{2}(M+m)v_2^2 + (M+m)gh = 0$$

$$v_2 = \sqrt{2gh}$$

$$\therefore v_i = \frac{m+M}{m} \sqrt{2gh}$$

- (b) Find the initial velocity of a 5.00 g bullet which is fired into a 1.25 kg block, if after the collision, the block with the bullet inside, rises to a maximum height of 22.0 cm.

Vind die aanvanklike snelheid van 'n 5.00 g koeël wat in 'n 1.25 kg blok gevuur word, indien na die botsing, die blok met koeël binne-in tot 'n maksimum hoogte van 22.0 cm styg. (2)

$$v_i = \frac{m + M}{m} \sqrt{2gh}$$

$$= \frac{5 \times 10^{-3} \text{ kg} + 1.25 \text{ kg}}{5 \times 10^{-3} \text{ kg}} \sqrt{2(9.8 \text{ m/s}^2)(22 \times 10^{-2} \text{ m})}$$

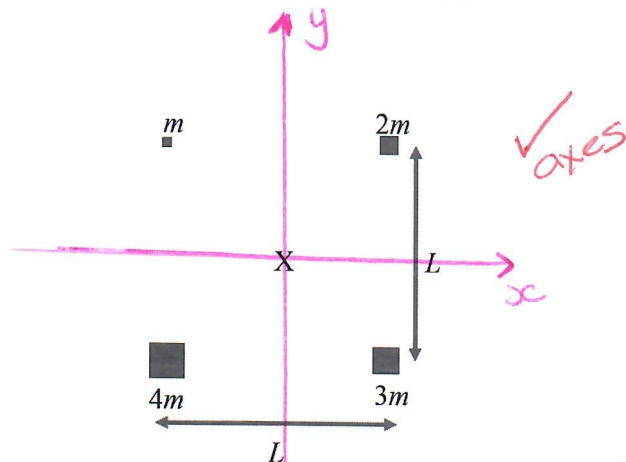
$$= 521 \text{ m/s}$$

7 Question 7 / Vraag 7

[7]

Masses m , $2m$, $3m$ and $4m$ are located at the corners of a square of side length L as shown in the figure. Locate the centre of mass of the system. Place the origin of your set of axes at the centre of the square, marked with X . Show your choice of axes clearly.

Massas m , $2m$, $3m$ en $4m$ word op die hoeke van 'n vierkant met sylengte L geplaas, soos aangedui in die figuur. Bepaal die massa-middelpunt van die stelsel. Plaas die oorsprong van jou assestelsel by die middelpunt van die vierkant, gemerk met X . Wys duidelik jou keuse van die assestelsel.



m	$\left(-\frac{L}{2}; \frac{L}{2}\right)$
$2m$	$\left(\frac{L}{2}; \frac{L}{2}\right)$
$3m$	$\left(\frac{L}{2}; -\frac{L}{2}\right)$
$4m$	$\left(-\frac{L}{2}; -\frac{L}{2}\right)$

$$x_{\text{com}} = \frac{1}{10m} \left(m \left(-\frac{L}{2}\right) + 2m \left(\frac{L}{2}\right) + 3m \left(\frac{L}{2}\right) + 4m \left(-\frac{L}{2}\right) \right)$$

$$= 0$$

$$y_{\text{com}} = \frac{1}{10m} \left(m \left(\frac{L}{2}\right) + 2m \left(\frac{L}{2}\right) + 3m \left(-\frac{L}{2}\right) + 4m \left(-\frac{L}{2}\right) \right)$$

$$= \frac{-4m}{10m} \left(\frac{L}{2}\right) = -\frac{1}{5}L$$

$$\underline{r}_{\text{com}} = \left(0\hat{i} - \frac{1}{5}L\hat{j}\right)$$

8 Question 8 / Vraag 8

[8]

A wheel starts to rotate from rest with a constant angular acceleration of $\alpha = 1.50 \text{ rad/s}^2$. The wheel undergoes this constant angular acceleration for $t = 20.5 \text{ s}$ and then continues to rotate at a constant angular velocity, ω .

'n Wiel begin roteer vanuit rus met 'n konstante hoekversnelling van $\alpha = 1.50 \text{ rad/s}^2$. Die wiel ondergaan hierdie konstante hoekversnelling vir $t = 20.5 \text{ s}$ en daarna roteer dit teen 'n konstante hoeksnelheid, ω .

- (a) What is the angular velocity after 20.5 s?
Wat is die hoeksnelheid na 20.5 s?

$$\omega_i = 0$$

$$\alpha = 1.50 \text{ rad/s}^2$$

(2)

$$\omega = \omega_i + \alpha t$$

$$= 0 + 1.50 \text{ rad/s}^2 (20.5 \text{ s})$$

$$= 30.75 \text{ rad/s}$$

$$= 30.8 \text{ rad/s}$$

- (b) After the wheel has been rotating for 60.0 s at the constant angular velocity from (a), what is the angular displacement of the wheel after 80.5 s?

Nadat die wiel vir 60.0 s geroteer het teen die konstante hoeksnelheid van (a), wat is die hoekverplasing van die wiel na 80.5 s

(6)

2 stages: $0 \leq t \leq 20.5 \text{ s}$ and $20.5 \text{ s} \leq t \leq 80.5 \text{ s}$

For $0 \leq t \leq 20.5 \text{ s}$ $\alpha = 1.50 \text{ rad/s}^2$

$$\Delta\theta_1 = \omega_i t + \frac{1}{2} \alpha t^2 = 0 + \frac{1}{2} (1.50 \text{ rad/s}^2) (20.5 \text{ s})^2$$

$$= 315 \text{ rad}$$

For $20.5 \text{ s} \leq t \leq 80.5 \text{ s}$ ($60 \text{ s} = \Delta t$); $\omega = 30.8 \text{ rad/s}$

$$\Delta\theta_2 = \omega t + \frac{1}{2} \alpha t^2$$

$$= 30.8 \text{ rad/s} (60 \text{ s}) + 0$$

$$= 1848 \text{ rad}$$

$$\Delta\theta_{\text{total}} = \Delta\theta_1 + \Delta\theta_2 = 2163 \text{ rad}$$

9 Question 9 / Vraag 9

[8]

The angular acceleration of a wheel is $\alpha = 6.0t^4 - 4.0t^2$, with α in radians per second-squared and t in seconds. At time $t = 0$, the wheel has an angular velocity of $+2.0$ rad/s and an angular position of $+1.0$ rad.

Die hoekversnelling van 'n wiel word gegee deur $\alpha = 6.0t^4 - 4.0t^2$, waar α in radiale per sekonde-kwadraat gemeet word en t in sekondes. By $t = 0$, het die wiel 'n hoeksnelheid van $+2.0$ rad/s en het 'n hoekposisie van $+1.0$ rad.

- (a) Write an expression for the angular velocity (rad/s) as a function of time (s).

Vind 'n uitdrukking vir die hoeksnelheid (rad/s) as 'n funksie van die tyd (s).

(4)

$$\omega = \int \alpha dt = \int (6.0t^4 - 4.0t^2) dt$$

$$= \frac{1}{5}(6.0)t^5 - \frac{1}{3}(4.0)t^3 + c$$

$$\text{At } t = 0, \omega = 2.0 \text{ rad/s}$$

$$\therefore c = 2.0 \text{ rad/s}$$

$$\omega = 1.2t^5 - 1.3t^3 + 2.0$$

- (b) Write an expression for the angular position (rad) as a function of time (s).

Vind 'n uitdrukking vir die hoekposisie (rad) as 'n funksie van die tyd (s).

(4)

$$\theta = \int \omega dt = \int (1.2t^5 - 1.3t^3 + 2.0) dt$$

$$= \frac{1}{6}(1.2)t^6 - \frac{1}{4}(1.3)t^4 + 2.0t + c'$$

$$\text{At } t = 0, \theta = +1 \text{ rad}$$

$$\therefore c' = 1 \text{ rad}$$

$$\theta = 0.2t^6 - 0.3t^4 + 2.0t + 1$$

$$= \frac{1}{5}t^6 - \frac{1}{3}t^4 + 2.0t + 1$$

END OF TEST / EINDE VAN TOETS